Electronic devices & Circuit Lab

EEL-224

Lab-07

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Experiment 07

RC Coupled Cascaded Amplifier

Objectives:

- The objective of this lab is to calculate the voltage gain and to observe frequency response of RC Coupled amplifier.

Apparatus:

1. DC power supply
2. Digital Multimeter
3. Dual-trace oscilloscope
4. Function Generator
5. Circuit breadboard
6. Transistor: Two Q2N2222
7. Resistors: 220 Ω,150 Ω, Two 15 kΩ, Four 3.9 kΩ, Two 4.7 kΩ, 3 kΩ
8. Capacitors: Two 2.2 μF, Three 10 μF

Introduction:

This is most popular type of coupling as it provides excellent audio fidelity.

A coupling capacitor is used to connect output of first stage to input of second stage. Resistances R1, R2 and $R_e$form biasing and stabilization network. Emitter bypass capacitor offers low reactance paths to signal coupling Capacitor transmits ac signal, blocks DC. Cascade stages amplify signal and overall gain is increased total gain is less than product of gains of individual stages. Thus, for more gain coupling is done and overall gain of two stages equals to A = A1*A2

A1 = voltage gain of first stage

A2 = voltage gain of second stage.

When ac signal is applied to the base of the transistor, its amplified output appears across the collector resistor Rc.It is given to the second stage for further amplification and signal appears with more strength. Frequency response curve is obtained by plotting a graph between frequency and gain in db .The gain is constant in mid frequency range and gain decreases on both sides of the mid frequency range. The gain decreases in the low frequency range due to coupling capacitor Cc and at high frequencies due to junction capacitance C_{be}

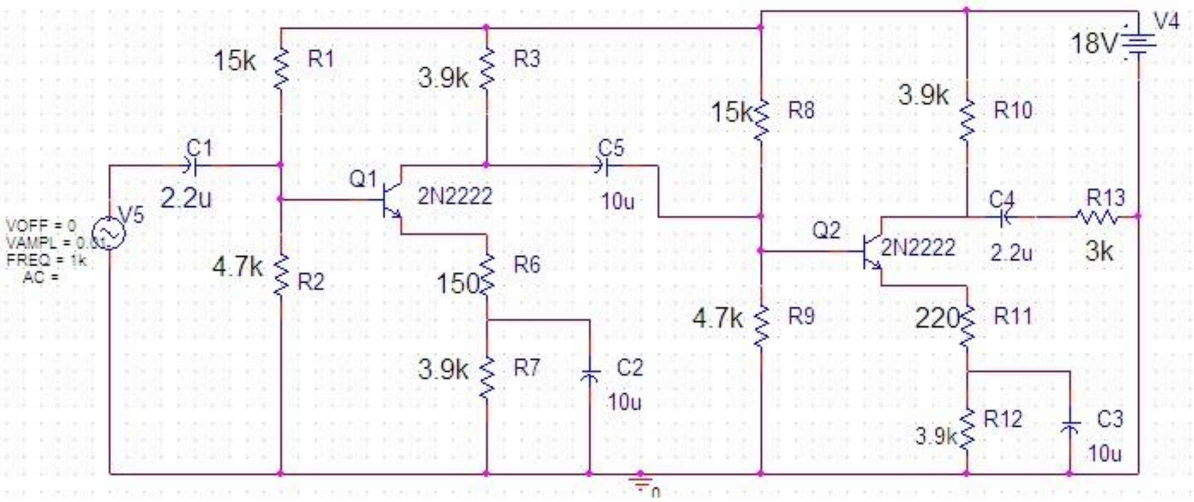
Theory:

An RC-coupled cascaded amplifier is a multi-stage transistor amplifier in which the output of one common-emitter stage is connected to the input of the next stage through a coupling capacitor and resistive network. This type of amplifier is widely used in audio-frequency applications because it provides good fidelity, stable performance, and significant overall gain. When an AC input signal is applied to the first transistor stage, it is amplified at the collector resistor and the resulting larger AC signal is passed to the second stage through a coupling capacitor. The purpose of this capacitor is to allow only the AC component to pass while blocking the DC bias of one stage from affecting the next. The resistors R1, R2, and RE form the voltage divider biasing network, which ensures that each transistor operates at a stable Q-point independent of transistor parameter variations. The emitter bypass capacitor provides a low-reactance path to AC, thereby increasing the AC gain of the amplifier.

When two stages are cascaded, the overall voltage gain becomes the product of the gains of the individual stages. However, practical loading effects such as the input resistance of the second stage drawing current from the first cause the overall gain to be slightly lower than the ideal value. The RC-coupled amplifier also exhibits a distinct frequency response. At low frequencies, the reactance of coupling and bypass capacitors becomes high, causing attenuation of the signal and reducing gain. In the mid-frequency region, these capacitors behave almost ideally, resulting in a flat and stable gain, which is the most useful operating region for audio signals. At high frequencies, internal transistor capacitances begin to dominate, reducing gain due to increased attenuation.

Overall, the RC-coupled cascaded amplifier provides high and stable gain over the mid-frequency range, making it ideal for audio amplification and general signal-conditioning applications. Its simple design, good frequency response, and ease of coupling multiple amplification stages make it one of the most commonly used amplifier configurations in analog electronics.

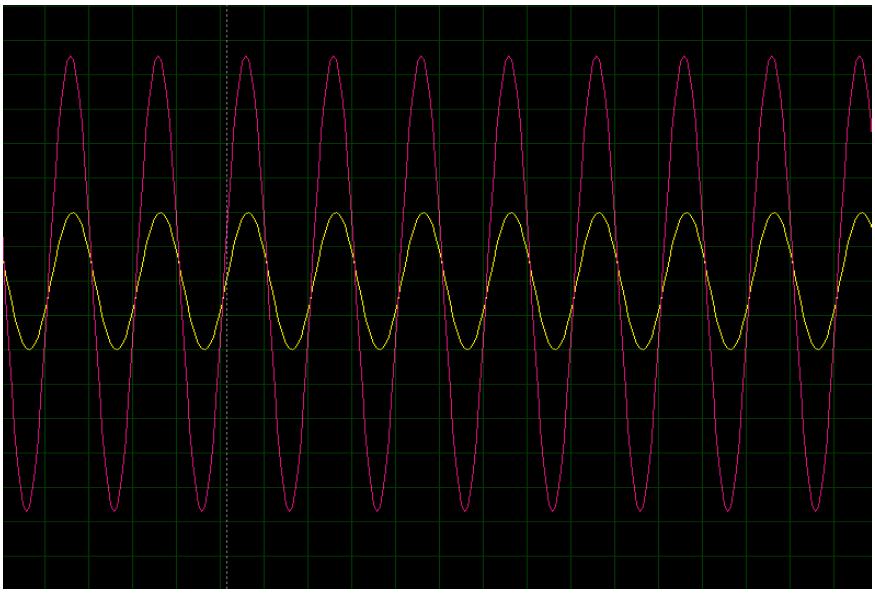
Circuit Diagram:



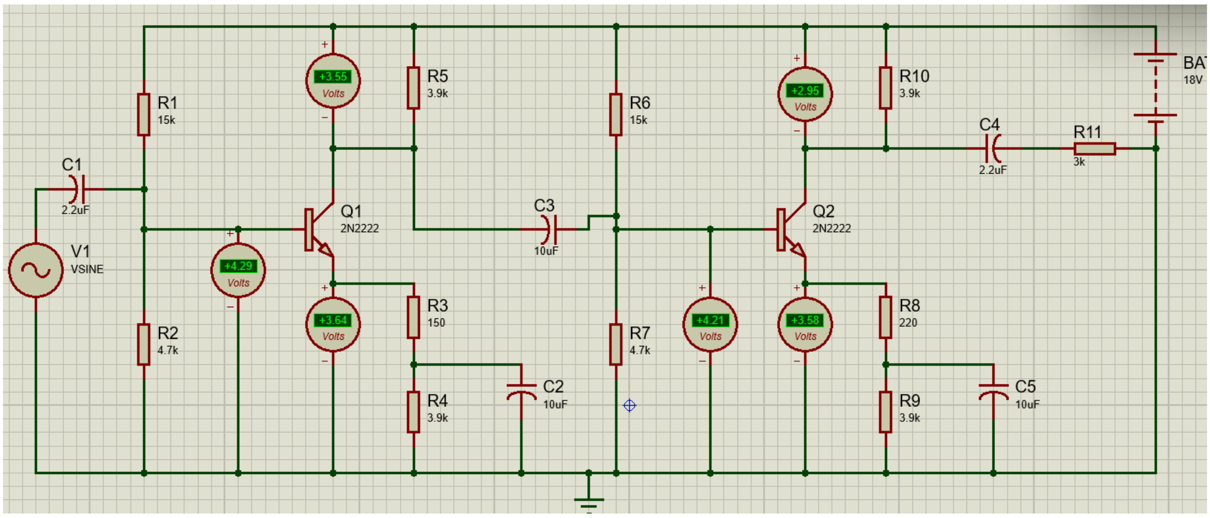
Procedure:

1. Apply input by using function generator to the circuit.
2. Observe the output waveform on CRO.
3. Measure the voltage at
   a. Output of first stage
   b. Output of second stage.
4. From the readings calculate voltage gain of first stage, second stage and overall gain of two stages. Disconnect second stage and then measure output voltage of first stage calculates voltage gain.
5. Compare it with voltage gain obtained when second stage was connected.
6. Note down various values of gain for different frequencies.
7. A graph is plotted between frequency and voltage gain.

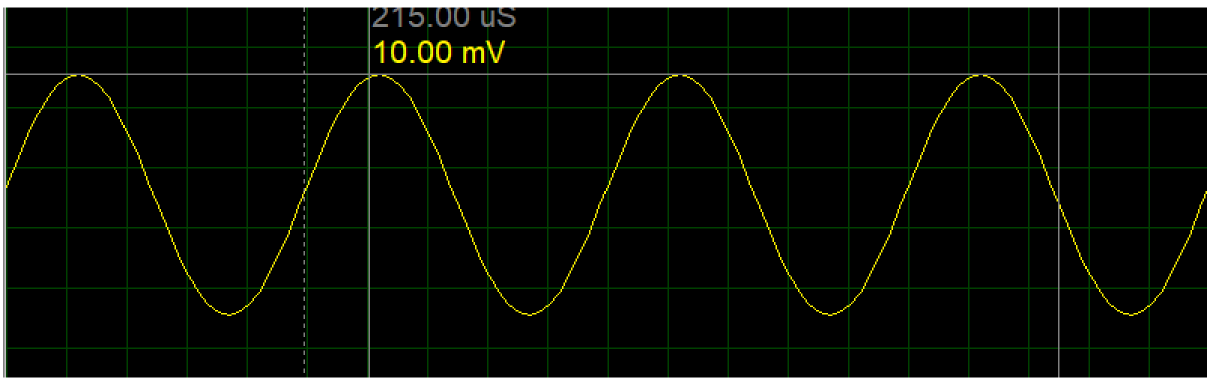
Graph:



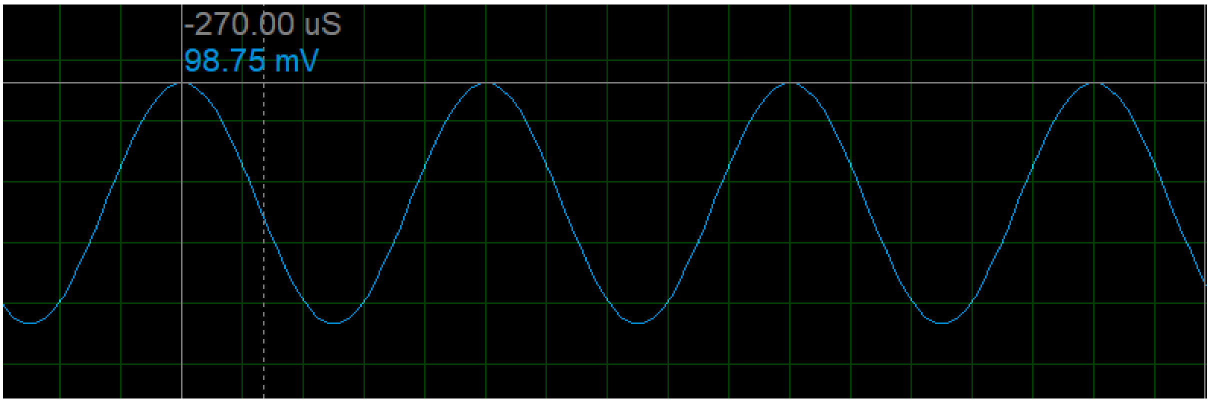
Simulations:



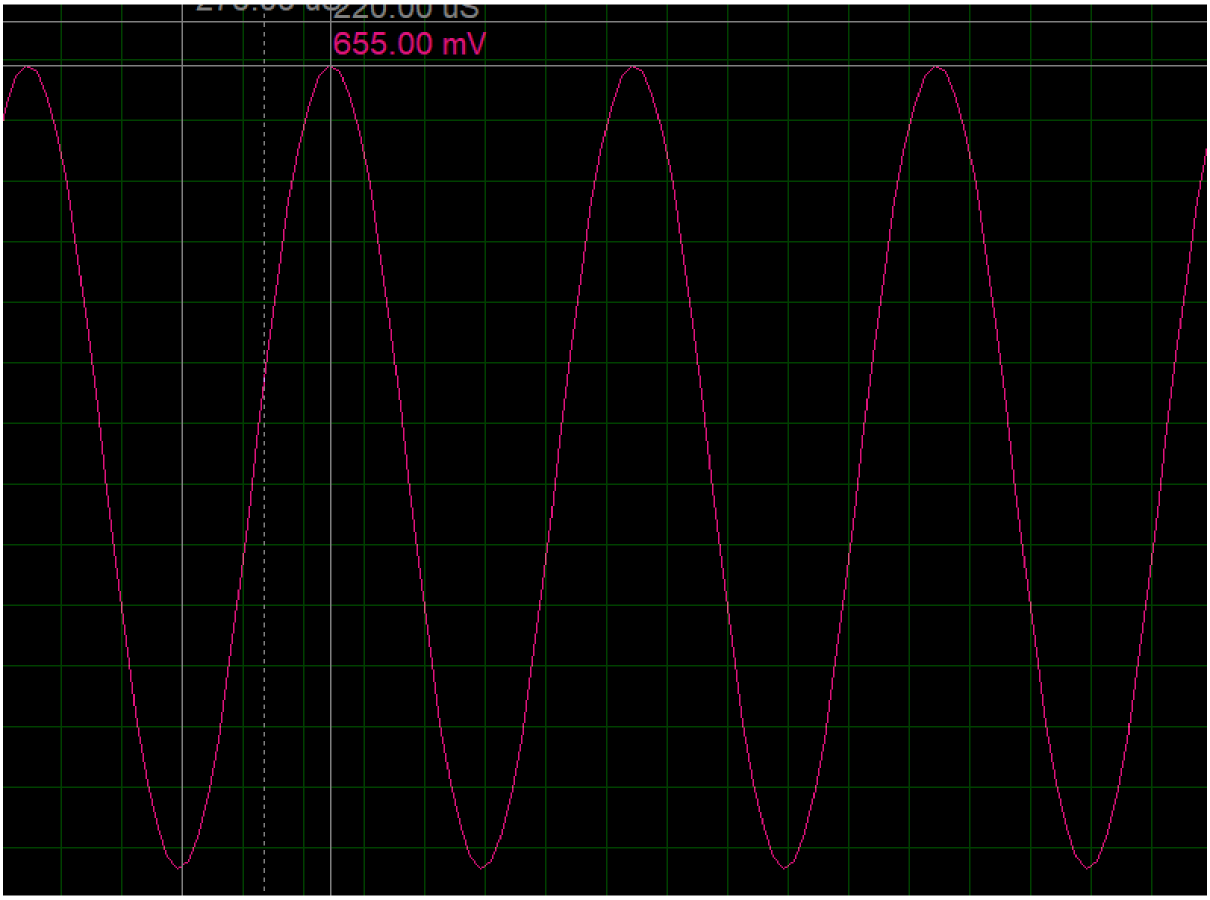
Simulation RC Coupled Cascaded Amplifier



Waveform 1: Input Waveform



Waveform 2: First Stage Output Waveform



Waveform 3: Second Stage Output Waveform

Measurement:

Parameter	Measured Value	Expected Value	% Error
Stage1			
V_B	4.28 V	4.20 V	1.91%
V_C	3.49 V	3.55 V	1.69%
V_E	3.63 V	3.58 V	1.40%
Gain Stage 1	-0.813	-0.820	-0.85%
Stage2			
V_B	4.27 V	4.22 V	1.81%
V_C	3.31 V	3.38 V	2.70%
V_E	3.62 V	3.57 V	1.40%
Gain Stage 2	-0.813	-0.820	-0.85%
Overall Gain	0.661	0.670	1.34%

Discussion:

In this experiment, a two-stage RC coupled amplifier was studied to observe voltage gain and frequency response. The first stage (Q1) amplified the input signal, producing a measured voltage gain of -0.813, which closely matched the expected gain of -0.820, resulting in only -0.85% error. The second stage (Q2) provided a similar voltage gain, and the overall cascaded gain was 0.661, very close to the expected 0.670 (1.34% error), demonstrating proper cascading effect. The measured base, collector, and emitter voltages for both stages were consistent with theoretical values, indicating correct transistor biasing and stable operation. Minor discrepancies in voltage and gain are likely due to component tolerances and measurement limitations. Overall, the experiment verified that cascading amplifier stages increases the overall voltage amplification while maintaining predictable performance. The RC coupling capacitors effectively transmitted AC signals between stages while blocking DC, ensuring proper biasing and signal amplification.

Advantages of the methods:

- Provides higher overall voltage gain by cascading multiple amplifier stages.
- Simple and easy to implement using standard resistors, capacitors, and transistors.

- RC coupling allows AC signals to pass between stages while blocking DC, maintaining proper biasing.
- Suitable for audio and low-frequency signal amplification.
- Easy to measure input and output voltages using standard lab instruments like DMM and CRO.

Disadvantages of methods:

- Voltage gain decreases at very low and very high frequencies due to coupling and bypass capacitors and transistor internal capacitances.
- Phase shift can occur if multiple stages are cascaded improperly.
- Component tolerances can cause small deviations in expected performance.
- Not suitable for very high-frequency applications (RF) due to RC coupling limitations.
- Requires careful biasing to ensure transistors operate in the active region; otherwise, distortion or improper amplification occurs.

Precautions:

1. Verify all circuit connections before turning on the power supply.
2. Ensure that the transistors are connected correctly (emitter, base, collector).
3. Use the recommended supply voltage to avoid damaging the transistors.
4. Keep the input signal small to prevent overdriving the amplifier.
5. Check the polarity of electrolytic capacitors before connecting them.
6. Turn off the power supply whenever making adjustments or changes to the circuit.
7. Handle wires and components gently to avoid short circuits or breakage.
8. Avoid placing the circuit near strong electromagnetic interference sources to prevent measurement errors.

Conclusion:

In this experiment, the operation of a two-stage RC coupled amplifier was successfully studied. The voltage gains of the individual stages and the overall cascaded amplifier were measured and found to closely match the expected theoretical values, with minimal errors. The RC coupling capacitors effectively transmitted AC signals between stages while blocking DC, ensuring proper biasing of both transistors. The frequency response demonstrated that gain remains relatively constant in the mid-frequency range but decreases at low and high frequencies, as expected. Overall, the experiment confirmed that RC coupling is an effective method for cascading amplifier stages to achieve higher voltage gain while maintaining signal integrity.